PRD: Offline USDT scam check for P2P traders

A cut-down MVP for the AI Yako hackathon. It checks addresses and chats on the device, offline, before a trader sends or releases USDT. A small receipt proof of concept comes second. Revised Thu 29 Oct 2026, 04:00 UTC.

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Summary and user

The user pastes or shares an address, a chat or a screenshot. The app checks it on the device and shows one level: Stop, Check more or No flags found. It shows the reason and the exact text that triggered it. Nothing leaves the device. After the first model download, it works in airplane mode.

Lead user	A P2P USDT seller in Nairobi who is paid by M-Pesa. Illustrative persona, not an interviewed user.
Moment of use	Before releasing USDT to a buyer, or before sending USDT to an address someone gave them.
Also works for	Sellers in Ghana (mobile money) and Nigeria (bank transfer). The address check and chat check don't depend on country.
User research	None done yet. To confirm with real traders.
Track	Offline AI apps (700 USDt) on AI Yako, DoraHacks 2452. Deadline Fri 30 Oct 2026, 14:00 UTC.

What the app never says

It never says an address, chat or receipt is safe. A frozen address is not proof of a scam. An address missing from the list proves nothing. A screenshot is not proof of payment.

Principle: code decides, model tags

Code decides

Fixed, tested code sets every risk level. The same input always gives the same level.

Model tags

The model only reads text and returns tags, each tied to the exact text that triggered it.

Add, never remove

A model tag can raise the level. It can never lower a level that code set, or remove a warning.

No guessing

When the app can't read or place something, it says Can't tell.

Core feature 1: Address check

1. **Input**

   Paste an address, or OCR it from a screenshot. The user picks the network they plan to send on: Tron (TRC-20) or an EVM chain.

2. **Validate**

   Tron: base58check checksum. EVM: `0x` plus 40 hex characters, and the EIP-55 checksum when the address has mixed case.

3. **Wrong network**

   Flags a Tron address when the user picked EVM, and the reverse. Code can't tell Ethereum from other EVM chains, since they share one address format.

4. **Frozen list**

   Checks a bundled offline copy of Tether's frozen-address list, about 11,900 Tron and Ethereum addresses, from the cryptoalert/usdt-blacklist-archive dataset. The file is stamped with the date it was taken.

5. **Look-alike**

   Compares the full address with saved contacts and past recipients. An exact match shows the contact's name. A shared prefix and suffix with a different middle is flagged as possible address poisoning. Prefix and suffix length to tune on test cases.

Result	Level	Wording
Invalid	Stop	This isn't a valid address. Check for a typo or a bad OCR read.

Wrong network	Stop	This address is for a different network than the one you picked.
On frozen list	Stop	Tether has frozen this address. That isn't proof of a scam, but funds sent there may be stuck.
Look-alike	Stop	This looks like an address you've used, but it isn't the same one.
Not on list	No flags found	Not on the frozen list as of the list date. This proves nothing about the address.

Pure code. No model.

Acceptance criteria: Address check

- [ ] A valid Tron address with one changed character returns Invalid.
- [ ] A lower-case EVM address passes format checks. A mixed-case one with a bad EIP-55 checksum returns Invalid.
- [ ] A Tron address checked with EVM picked returns Wrong network, and the reverse.
- [ ] An address from the bundled frozen list returns On frozen list and Stop.
- [ ] An address sharing prefix and suffix with a saved contact, but not the middle, returns Look-alike.
- [ ] An exact match with a saved contact shows that contact's name.
- [ ] No result ever uses the word safe.
- [ ] All of the above pass as unit tests in the shared core package, with the network off.

Core feature 2: Chat check

1. **Input**
   Paste chat text, or OCR a screenshot with QVAC OCR.
2. **Rules**
   Fixed phrase rules in English and Swahili find hard signals, such as a request for a PIN, seed phrase or OTP.
3. **Tag**
   One small QVAC LLM returns tags as fixed JSON: pressure, impersonation, moving off-platform

One small QVAC LLM returns tags as fixed JSON: pressure, impersonation, moving off platform, fake support. Each tag carries the exact triggering text.

4 **Addresses**

Any address in the chat goes through the address check.

5 **Decide**

Code maps rules, tags and address results to one level, with every reason and its text shown.

Signal	Level
Rule hit: asks for PIN, seed phrase or OTP	Stop
Any address result at Stop	Stop
Two or more different model tags	Stop
One model tag	Check more
Nothing found	No flags found

This is a starting table. Tune it on the dev split only. The model is to confirm on the phone, picked by speed and output quality. Whether it handles Swahili directly, or needs a translation step, is also to confirm.

Acceptance criteria: Chat check

- [ ] A pasted chat asking for a seed phrase returns Stop and shows that sentence.
- [ ] An OCR'd screenshot of the same chat gives the same level.
- [ ] Each tag shown on the result card quotes the exact text that triggered it.
- [ ] A unit test feeds a model output with no tags into a Stop case. The level stays Stop.
- [ ] A chat with an address from the frozen list returns Stop.
- [ ] The check runs in airplane mode on the target device.
- [ ] A small labelled chat set is split into dev and test before tuning. The README reports measured recall and false positives from test only, or says not yet measured.

Receipts: thin proof of concept

One screen that shows flag-and-verify works on images too. It is not a headline feature. Kinga does receipt checking better and we don't compete there.

Reuse detection · In scope

SHA-256 of the file, a perceptual hash, and a local ledger of transaction IDs. A repeat returns Reused.

Fields vs pasted SMS · If time

OCR'd amount and transaction ID compared with SMS text the user pastes. Exact match only.

Format files · Placeholder

Placeholder files for Ghana, Kenya and Nigeria. Formats TBD until we have real, consented samples.

Unknown provider · In scope

Returns Can't tell. The app does not guess.

Acceptance criteria: Receipts

- [ ] The same receipt shared twice returns Reused the second time.
- [ ] A re-saved copy of a known receipt returns Reused through the perceptual hash.
- [ ] A receipt from an unknown provider returns Can't tell.
- [ ] No receipt result reads as verified or safe.
- [ ] If built: an edited amount that disagrees with the pasted SMS returns Mismatch.

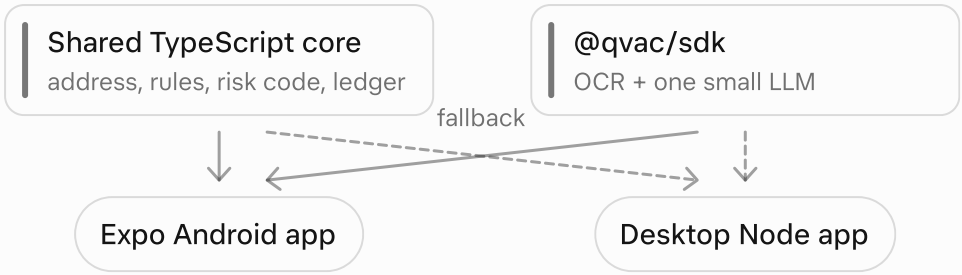
Offline-proof script

A script, like Kinga's `scripts/offline-proof.ts`, runs the address, chat and receipt checks end to end and logs every outbound network attempt. It runs in airplane mode and appears in the demo video.

- [ ] A full run logs zero outbound network calls.
- [ ] The script and its output are in the public repo, with steps to rerun it.
- [ ] The demo video shows airplane mode on and the script's zero result.

Platform risk and fallback

QVAC on Android needs a physical arm64 phone on Android 12 or later, minSdk 29. There's no emulator. The Expo plugin is the riskiest part of the build, so it's tested first. The submission doesn't require an APK.



Go/no-go	Thu 07:00 UTC. A QVAC model loads and answers a prompt on the phone.
Core package	All checks live in a platform-independent TypeScript package with its own tests. The phone and desktop apps are thin shells.

If	Then
The Android build fails by the gate	Ship the same core as a desktop Node app with `@qvac/sdk`. Record the same demo flow in airplane mode.
The model is too slow on the phone	Try a smaller model. If still too slow, the phone runs the address check and rules only, and says so.
No arm64 phone is available	Desktop only, declared honestly in the device and network declaration.

Cut list and competition

Cut to future work	Why
Vision-model receipt extraction	Too slow to get right by Friday. Kinga leads here.
Provider formats beyond placeholders	Need real samples first.
Balance-chain check	Needs full SMS history and verified formats.
Whisper voice notes	Not core to the two checks.
Translation beyond Swahili	Languages to confirm. Twi is unverified.
WDK test-wallet step	Bonus track only. Not core.

Project	What it is	How we differ
Kinga (Al Yako, same track)	Desktop web app for Kenya. Checks M-Pesa receipts	Phone-first, frozen list, address check. Its receipt checking is

	against imported SMS. About 4 GB of models and a GPU PC.	stronger.
Custos	Browser app. Chat OCR, Spanish and English phrases, look-alike on first and last 6 characters, Tron testnet.	Checksum validation, frozen list, full-address compare, Swahili.
Scam Shield Offline (AI Yako)	Desktop Node app. Pasted text, 10 weighted rules, a model explains the top flag.	Same rules-first idea. We add addresses, OCR and a phone build.

Build plan to Fri 30 Oct, 14:00 UTC

When (UTC)	Milestone	Done when
Thu 04:00–05:00	Repo, core TS package, test runner	Empty test passes in CI or locally
Thu 05:00–07:00	Expo app with QVAC plugin on the phone	Go/no-go at 07:00: a model answers on the phone
Thu 07:00–09:00	Address check in core: validators, frozen list bundle, look-alike	Address criteria pass as unit tests
Thu 09:00–10:00	Address screen on phone, or desktop if no-go	Paste an address, see a level and reason
Thu 10:00–12:00	Chat check: rules, OCR, one LLM tagger, risk code	Model-can't-lower test passes
Thu 12:00–13:00	Chat screen with reasons and quoted text	Chat criteria pass on a few examples
Thu 13:00–14:00	Break	
Thu 14:00–15:00	Offline-proof script	Zero outbound calls logged
Thu 15:00–17:00	Receipt screen: hashes, ledger, placeholders, Can't tell	Receipt criteria pass
Thu 17:00–18:00	If time: OCR fields vs pasted SMS	Mismatch case passes, or cut
Thu 18:00–19:00	Label small chat set, dev/test split, run once	Numbers recorded from test only
Thu 19:00–Fri 05:00	Rest	

Fri 05:00–06:00	Fix list, full airplane-mode run on the device	All three checks give results offline
Fri 06:00–08:00	README, QVAC details, device and network declaration, honest limits	A fresh clone builds from the README
Fri 08:00–10:00	Record the offline demo video	Airplane mode and the script result on screen
Fri 10:00–11:00	Upload video. APK only if Android passed the gate	Public link works
Fri 11:00–12:30	Submit on DoraHacks	All checklist items attached
Fri 12:30–14:00	Buffer	

Solo. Times are UTC, which is also Accra time. About 21 working hours, with breaks and sleep.

Submission checklist

Status as of Thu 29 Oct 2026, 04:00 UTC: nothing built yet.

- [ ] Public repo with the core package, app shell and tests.
- [ ] Offline demo video recorded in airplane mode.
- [ ] QVAC details: which SDK functions and models, and where they run.
- [ ] Device and network declaration: device model, OS, and the one-time model download.
- [ ] README with honest limits and only measured numbers.

Open questions

1. Is this solo, or is there a team? How many hours can each person give?
2. Is there an arm64 Android 12+ phone to test on?
3. Is there an existing repo, or do we start fresh?
4. Which small QVAC LLM runs fast enough on that phone for tagging?
5. What licence and update date does the frozen-list dataset carry?
6. Will organisers accept a one-time model download as offline?

References

- Kinga repository
  github.com
- Custos repository
  github.com
- Scam Shield Offline repository
  github.com
- QVAC OCR docs
  docs.qvac.tether.io